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E 460 SUBSTATION CONTROL SYSTEM

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E-460 SUBSTATION CONTROL SYSTEM

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of substation control system in accordance with Part 4, Subsection 1.2 Summary of Work and bid drawings for the extended substation and the existing diameter 5.

Remote Terminal Unit (RTU) shall be interface with DCS combined cycle power plant for data communication.

The control system for the existing substation is available. The Contractor shall provide or extent (whichever applicable) a microprocessor-based substation control system with enough capacity to control the extended part of the substation and the existing diameter 5. The control system is described under Technical Specification I-005 and bid drawings

The substation equipment will be controlled from the substation control mimic panel and from VDU located in the substation control room and from Java Control Center (Load Dispatch Center) via SCADA system. Selector switch with the following positions shall be provided:

Local – Remote (Substation) – VDU – SCADA, and all positions are referring to the substation equipment. The Contractor is responsible for providing and matching the connection to the existing substation control to realize a complete and comprehensive control system.

The Contractor shall provide all necessary interface for SCADA and cabling as described in Technical Specification E-400 and according to PLN Standard requirements.

The SCADA system interface should be based on the latest PLN Standard for load frequency control with provision to incorporate automatic generation control in the future.

The Contractor's control system shall be subject to the Owner's review and approval.

The substation shall also be monitored from the console (VDU) in the power plant central control room.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Section 2.5 Codes and Standards and the following specific standards:

2.1 Indonesian National Standards

PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)
SPLN : Standard Perusahaan Umum Listrik Negara.
DITJEN LPE : Direktorat Jendral Listrik dan Pemanfaatan Energi mengenai : Aturan Jaringan Jawa-Madura-Bali

2.2 Institute of Electrical and Electronics Engineers

ANSI/IEEE C37.1 : Definition, Specification, and Analysis of Systems Used for Supervisory Control, Data Acquisition, and Automatic Control.

3. OPERATING AND DESIGN CONDITIONS

The substation controls shall be designed for operation indoors or outdoors as required for a site located in a tropical climate. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions.

4. DETAILED REQUIREMENTS

4.1 General

The following definitions shall be used for clarifying the manual control functions specified hereunder.

- (1) “Local” means control activity initiated at the primary equipment box or the marshalling cabinet associated with the equipment.
- (2) “Remote” means control activity initiated from mimic/control panel at the substation control room.
- (3) “VDU” means control activity initiated from the functional keyboard with VDU at the substation control room.

- (4) “SCADA” means control activity initiated at the Java Control Center (JCC) through the existing fiber optic carrier.
- (5) “500 kV Generator Circuit Breaker” means 500 kV circuit breaker at 500 kV switchyard associated with generator line output from generator transformer HV winding terminal.

The control system shall be designed to provide a good operational flexibility and safety interlocks necessary to protect personnel and the equipment in an HV substation.

The manual control of the substation equipment, in general, shall be executed from a substation control panel located in the substation control room. The control system shall also be designed to permit a remote control of the substation equipment from JCC.

For the purpose of maintenance, testing, etc., the control system shall be designed to provide local control facilities and indications at the respective equipment.

4.2 Functional Control Requirements

4.2.1 Interlocking

4.2.1.1 General

- (1) Operational electrical interlocking shall be provided to assure that the equipment will perform in a safe manner. Such interlocks shall be achieved by hardwired electrical means.
- (2) Maintenance interlocking shall be provided to render equipment or sections of substation safe for access and maintenance by personnel. Such interlock shall be achieved by mechanical and electrical interlock.
- (3) Where complete and safe interlocking is impossible a prominent warning notice shall be posted near the operating points of this device.
- (4) Contractor shall ensure that safe operation and maintenance interlocking is provided. Interlocking specified herein is considered minimum philosophy and shall be verified by the Contractor and enhanced as required

- (5) Sufficient spare contacts shall be provided.

4.2.1.2 Disconnecting Switches

- (1) Interlocking shall be provided such that the disconnecting switches for a circuit breaker isolation cannot be operated unless the circuit breaker is in open position.
- (2) Interlocking shall be provided on each generator transformer feed disconnecting switches such that the disconnecting switch cannot be operated unless all circuit breakers at that voltage level associated with the generator transformer are open.
- (3) Each 500 kV disconnecting switch at the new diameters 500 kV switchyard shall be interlocked to prevent its operation if any one (1) of the two (2) associated 500 kV circuit breakers is closed.
- (4) Interlocking shall be provided on all disconnecting switches to prevent closing when the grounding switches on either side of the related disconnecting switches are closed.

4.2.1.3 Grounding Switches

- (1) For interlocking purposes, it is assumed that operating personnel are adjacent to switch, therefore, all interlocking must have personnel safety in mind.
- (2) The grounding switch shall not be operable unless all disconnecting switches and/or circuit breakers which isolate the section to be grounded are open.
- (3) The grounding switch shall not be operable unless the remote-local selector switches at the associated disconnecting switches and circuit breakers are in the “local” position.
- (4) The grounding switches for generator transformer shall not be operable unless generator circuit breakers are open.

4.2.1.4 Circuit Breakers

- (1) Each circuit breakers shall be interlocked such that it cannot be closed when the grounding switch on either side is in closed position.

- (2) 500 kV generator circuit breakers will require interlocking with the power plant control systems.

All disconnecting switches have to be interlock to the relevant circuit breakers.

- (3) Protective relaying and synchronizing interlock are specified in the respective sections of the Bid Document. The Contractor shall provide these and other interlocks necessary for a safe operation.

4.2.2 Synchronizing

4.2.2.1 General

All 500 kV circuit breaker control shall be provided with synchronizing scheme as described hereinafter

- (1) Switching of 500 kV circuit breaker is only possible via the existing synchro-check relay system.
- (2) Switching of the generator circuit breaker is only possible via the power plant synchronizing system.

One (1) synchronizing system is already existing. The Contractor shall add any equipment required for the proper integration to the existing system. The following is to inform the Contractor of the requirement.

- (1) The synchronizing indication shall utilize existing equipment installed in the substation control room. The indication shall consist of synchronizing indicator, incoming and running volt, along with system and differential frequency, suitable for continuous operation.
- (2) The synchronizing scheme shall be interlocked with manual control of the breakers to assure that the circuit to be tied are within the proper limits of frequency, phase, and voltage level.
- (3) The synchronizing check output shall be provided with links or switches to make choice of which functions are to be used in control of an individual circuit breakers.

- (4) For circuit breakers connected to a bus, the “incoming voltage” is defined as the generator transformer voltage and the “running voltage” is the respective bus voltage.
- (5) Synchronizing voltage shall be phase to ground voltage on phase “S” and all equipment shall be suitable for 80% to 120% of the nominal voltage.
- (6) All voltage circuits shall have instrument-isolating transformers to prevent circulating ground current in voltage transformer secondary from affecting the synchronizing circuits. Also the connections shall be such that paralleling of voltage transformers secondary is prevented such that during component failure the system is fail-safe.

4.2.2.2 Synchronizing Check Relays

- (1) A synchronizing check (SC) shall include the following parameter check before permission to close is given to a breaker:
 - a. A phase angle check between incoming and running voltage shall be checked with a maximum angle difference adjustable between 20° and 45° to either side of exact synchronizing.
 - b. The maximum differential frequency (slip) between incoming and running voltage shall be checked, and will be adjustable from 0.01 to 0.05 Hz, the setting being based on breaker closing time such that closure of the circuit breaker contacts at virtually 0° phase angle difference.
 - c. Voltage of both incoming and running sides shall be checked and will have an adjustable minimum permissive range + 10% to -20% of nominal.
- (2) Dead bus voltage check (DBC) shall ensure that the running voltage is at virtual zero, and the incoming voltage is within the permissible voltage range for closing under synchronism check.
- (3) Dead line voltage check (DLC) shall ensure that the incoming voltage is at virtual zero, and the running voltage is within the permissible voltage range for closing under synchronism check.

- (4) A key interlocked SC bypass switch shall be provided to allow use of DBC and/or DLC manual closing.

4.2.2.3 Circuit Breaker Synchronizing Permissive Interlocks

The synchronizing scheme shall be arranged to provide the following interlocks:

- (1) All 500 kV circuit breakers shall have key interlocked synchronizing soft selector switches such that only one (1) circuit breaker can be synchronized at a time at any one (1) station. The switch shall have two (2) positions “off” and “manual synchronized”. The “manual synchronized” position shall have contacts to energize the respective incoming and running voltage instrument buses from the appropriate voltage transformers and to prepare the circuit breaker for closure by enabling the synchronizing check circuit in the closing control. The permission to close the circuit breaker shall be given when SC, DBC, or DLC conditions exist.
- (2) If a generator disconnecting switch is open, or a generator is already connected to the system, the generator circuit breaker(s) shall be synchronized as per Clause 4.2.2.3(1).

4.2.3 Manual Control

In general, the manual control of the substation equipment shall be as described in Clause 4.1. The following are the additional requirements for the manual control:

4.2.3.1 Local Control

The local control cabinet for each remote controlled equipment shall be provided with a key locked for local, remote, VDU and SCADA selector switch. Normally this selector switch will be in the SCADA position and under this condition, control from the local cabinet shall be inhibited. For the local operation of an equipment, during maintenance, testing, etc., the selector switch will be turned to the local position. Under this condition, and only after the equipment has been isolated from the system (de-energized), the operation of the equipment shall only be permitted from the local cabinet.

4.2.3.2 SCADA Control

All circuit breakers and disconnecting switches shall also be wired for the SCADA control, subject to the specified interlocks. For each SCADA controlled

equipment, a selector switch is provided on each respective control panel.

4.3 Local Control Panels and Cabinets

4.3.1 General

All the local control panels supplied with the equipment shall meet the existing control panels specifications.

4.3.2 Breaker Bay

All necessary controls, indications, measuring indications, alarms, etc. shall be provided on the VDU and control panel to operate all the equipment in that bay.

All controls for one (1) bay of circuit breakers, isolating disconnecting switches and their associated equipment and operating alarms, branch protection alarms, indications of position or status, indication of measurements and any necessary auxiliaries bus diagram of the bay for easy operator understanding shall be included on the VDU.

Generator indicator metering shall consist of an ammeter for each phase, a Watt, a Var, and a Volt indication etc according to PLN standards.

4.3.3 Local Control Panels

The local control panels provided with each equipment shall include all control and indication devices necessary to meet the specified requirements as per Technical Specification I-002 and to permit proper maintenance and testing. The panels shall be sized liberally to accommodate all the equipment, arranged for an easy operation and maintenance.

The substation control system shall be designed for remote control of the substation from the substation control room and SCADA including all functions, alarms and indications necessary for operation.

The control system shall have full functional capability of receiving, acknowledging and resetting the annunciator alarm.

5. PERFORMANCE TESTING AND GUARANTEE

Component, wiring and other generic panel testing requirements are covered in Technical Specification I-002.

The control system shall be assembled, and all hardware's inputs and outputs shall be tested successfully in accordance with the Specification to the satisfaction of the Owner, and all modifications requested by the Owner shall be implemented without any cost to the Owner.

All components after assembly shall be energized and "burned in" for a minimum of 90 days to minimize "infant mortality" problems with components.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the substation control system shall conform to the requirements included in Part 4, Subsection 2.7.

The quality assurance/quality control procedures specified above shall include, but not be limited to the, followings:

- (1) All routine tests specified in Clause 5 of this Specification.
- (2) All structural and control design activities, calculations and document preparation.

The quality assurance/quality control documentations as specified above shall include, but not be limited to, the followings:

- (1) Shop Test Reports.
- (2) All other documentations required by Clause 5 test and as required by applicable codes and standards.

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